Response to Reviewers

Nanopore-only telomere-to-telomere assembly of diploid human genomes with haplotype-resolved methylation and 3D chromatin maps

Gross et al. : Manuscript for Nature Communications

We thank the editor and all four reviewers for their careful and constructive assessment of our manuscript. The reviewers' comments have substantially improved the work, by prompting an independent, reference-based evaluation of assembly accuracy against the GIAB HG002 benchmark, the addition of a methylation-based parent-of-origin assignment step, and a more thorough characterization of the residual assembly fragmentation. Below we respond to each comment in turn. Reviewer comments are shown in italics, our responses in green. Changes in the manuscript are marked in red color; line numbers refer to the revised manuscript.

General response

Revised quality values (QV)

In the previous version of our manuscript, we reported consensus accuracy from reference-free k-mer comparison (*Merqury*), with a median QV of 51. In this revision we added the HG002-based evaluation which confirmed that these values overestimate the true accuracy. K-mer QV assumes that k-mers missing from the read set are assembly errors, but in low-complexity and repetitive sequence the same errors recur in most reads, thus they are not detected. We now report HG002-based QV as the primary accuracy metric. QV values for the newly assembled individuals are reported as before using *Merqury* values. We state in the manuscript that the two metrics are not directly comparable in absolute terms. The HG002 benchmark is described or discussed in Methods (lines 532 – 558), Results (189 - 211) and Discussion (318 - 323). HG002-based QV values are now presented in the abstract.

A note on cohort size

Throughout the manuscript, the cohort comprises 23 individuals (19 unrelated participants and, for two of these, both parents). In addition, we assembled the HG002 cell line as an independent benchmark. HG002 is not part of the cohort and is described throughout as a benchmark assembly rather than as a study sample. Where it is included in a specific analysis (for example, the contiguity analysis), this is stated explicitly. Sample counts in figures include the 23 samples plus the HG002 benchmark.

Reviewer #1

We thank the reviewer for the constructive comments on code availability and documentation.

Code availability: release tags

> *Please make releases of the github repo that mark important milestones, e.g., the version of the code that you used to run the analyses described in the manuscript.*

Done. We have added a release tag (v1.0) marking the code version used for the analyses in the manuscript. We will add a final release tag (v1.1) for the pipeline when the paper is accepted.

Code availability: license

> *Please add a license file, preferably a permissive open source license such as the MIT.*

Done. An MIT License file has been added to the repository.

Documentation

> *Your main readme offers some pointers to the individual parts of the analysis, which I appreciate very much, but some of that presumably detailed information seems rudimentary at best (in particular the last point about runtimes, see my comment above). Please revise or remove things that you do not consider important.*

Done. We have revised the documentation, removed dead links and internal notes, and added instructions for running the pipeline.

Step-by-step pipeline instructions

> *Your main readme should ideally also contain a brief step-by-step instruction on how to run the Nextflow pipeline; at least for the main part, i.e. the assembly itself, which is very relevant for other groups who are interested in using your approach as well.*

Done. Step-by-step instructions for running the assembly pipeline have been added to the README.

Reviewer #2

We thank the reviewer for the positive assessment of the manuscript and for the detailed and constructive comments.

Major comment 1: HERRO correction and residual indels

> *My biggest concern is around HERRO correction and resultant indels. The authors note a modest increase in 1–5 bp indels, likely reflecting residual homopolymer-associated errors. This is important because small indel accuracy remains a concern for ONT-only assemblies. The manuscript would be strengthened by quantifying these errors more directly: indel rate by homopolymer length, coding sequence impact, overlap with low-complexity sequence, and comparison to HPRC or HG002 benchmark regions.*

Done. We added a detailed analysis of indel rates by homopolymer length and compared indel rates in coding-sequence vs. low-complexity regions using the newly added HG002 benchmark (Supplementary Fig. 13, 14; Result lines 189 – 205; Method lines 532 - 558).

We agree this point is central to the manuscript. Instead of inferring the error profile from the variant calls, we now measure it against an independent benchmark. We generated new data by sequencing the HG002 cell line with our standard recipe (three UL and one Pore-C flow cell), assembled it with the same pipeline, and compared it to the GIAB HG002v1.1 benchmark using *GQC*, classifying errors by sequence context with the GIAB v4.2 stratifications.

Indel rate by homopolymer length. Accuracy in mononucleotide runs drops with run length (Supplementary Fig. 14D). Runs below 10 bp are resolved reliably, accuracy falls steeply between 10 and 25 bp, while runs above 25 bp are essentially unresolved. The overall error rate in mononucleotide runs is 20.3%, dominated by single-base increases/decreases (about 29 1bp-deletions and 26 1bp-insertions per Mb). 2pb-events are fourfold rarer and discrepancies of length 4 or more are negligible. Out HG002-benchmark therefore shows that short indels in long homopolymer runs remain the primary issue for Nanopore-only genome assembly. This could be resolved in the future using the ONT APK polishing kit for HG002 (not tested in this study).

Coding sequence. The complete coding sequence contains 56 errors in 70.3 Mb (QV 61.0), 41 of them indels and 15 substitutions.

Low complexity vs. non-repetitive regions. Errors are strongly concentrated in low complexity / repetitive regions: 89.5% of all errors are found in in homopolymers, 91.5% in homopolymers and tandem repeats together, and 99.8% in any repeat or low-complexity region. In comparison, non-repetitive sequence showed only 781 errors across 4.83 Gb (QV 67.9). Indels make up 97.4% of all errors and substitutions 2.6%. Across the whole genome indels make up 97.4% of all errors and substitutions 2.6%.

Choice of region stratification. We used the GIAB v4.2 region stratification, which supersedes the earlier high/low-confidence region definition. GIAB v4.2 region stratification defines seven categories, which are in our opinion more informative than a simple split in high and low confidence regions.

We have added the new procedures and results to the sections Results (line 196 - 205) and Methods (see new paragraph “Independent evaluation against the GIAB Q100 HG002 benchmark”).

Major comment 2 : reproducibility of four flow cells per genome

> *I am a bit confused as to how reliable/reproducible 4 flowcells/genome is. As a reader, a clear table for every sample listing: number of UL flow cells generated, number used for assembly, number of Pore-C flow cells generated, number used for assembly/phasing, total bases, read N50, fraction of reads >100 kb, final assembly classification, QV, number of T2T contigs, number of T2T scaffolds, and centromeres resolved.*

Done. We added a comprehensive per-sample sequencing and assembly summary table (Supplementary Table 2). Manuscript amended (lines 116-121).

Supplementary Table 2 provides a per-sample overview of the flow cells generated and used for each assembly, together with sequencing yield, read N50, fraction of reads >100 kb, assembly classification, T2T contig and scaffold counts, and resolved centromeres.

For all assemblies, a single Pore-C flow cell was used, which is sufficient for phasing and scaffolding. A second Pore-C flow cell was sequenced for every sample for deep chromatin contact analyses, with further Pore-C flowcells added for selected samples for generating high-resolution contact maps. None of these additional PoreC flowcell data were used for assembly.

The target of four flow cells per assembly was met for all later samples (T2T05 - T2T19 except T2T10, i.e. 14 samples), for which library preparation and sequencing had been optimized. For the first batch of samples (T00 - T04) we still required more FC to reach a comparable number of reads longer than 100 kb. We therefore consider the claim of four flow cells per genome to be reliable and reproducible using our optimized protocol. We have added these details to the manuscript (lines 116-121).

We did not include a QV column in this table, as we now report HG002-based QV values, which we consider more accurate than the reference-free k-mer estimates used in the previous version of our manuscript. These values are reported in Supplementary Figures 13 and 14.

Minor comments

> *The Discussion mentions rare disease, cancer predisposition, and clinical translation. This is reasonable, but the manuscript is not primarily a clinical validation study. I would soften clinical language unless the authors add more evidence about clinically relevant variant detection.*

We agree. Clinical language has been softened or removed in the Introduction and Discussion. Clinical applications are described as prospects enabled by the workflow rather than as demonstrated outcomes.

> *Please be consistent about the use of the terms ultra-long reads, UL reads, or ULR.*

Done. Terminology harmonized to “UL reads” throughout the manuscript.

We have standardized the terminology to "ultra-long (UL) reads" at first use in Abstract, Graphical Abstract and Introduction and "UL reads" thereafter. We removed the variant "ULR" throughout.

> *In the intro the authors mention "we assembled 31 of 46 chromosomes gaplessly (Sample T2T17)", why did you choose this sample for the intro? Why not report an average across all samples?*

Agreed. We replaced the single best-case sample with the cohort median and range (line 103).

We have replaced the results for T2T17 with the median number of gaplessly assembled chromosomes across the cohort (16.5 of 46; range 0–31) and report the per-sample range and the best case in the Results.

> *The manuscript might benefit from a cost/labor comparison between this workflow and typical HiFi + ONT + Hi-C/Strand-seq/trio workflows.*

Done. We added a workflow comparison to the Discussion (299 – 306). Compute requirements are reported in Supplementary Fig. 4 and briefly listed in Results (lines 135 - 139).

We agree that the advantage of using a single technology approach should be made clearer, and we now address this in the Discussion (lines 299 - 306). We have deliberately not included monetary cost estimates as reagent, flow-cell, and labor costs vary strongly between institutions (due to different purchase discounts and cost per working hour), and are rapidly changing. Instead, we compare the workflows based on the estimated hands-on working time and instrument requirements. While our approach uses a single sequencing platform (ONT PromethION) with two library types, a combined HiFi + ONT-UL + Hi-C/Strand-seq workflow requires two to three sequencing platforms and three library preparations. Trios on the other hand require triple sequencing cost and hands-on work. Hence, the reduction in sequencing platforms to be bought/maintained and reduced hands-on time are the main measurable advantage.

> *Fig 2A, the overlap of dots makes it hard to appreciate; make dots smaller and separate them slightly.*

Done. Dot size reduced and a group offset added in Fig. 2A.

> *Fig S1, APK is not defined.*

Done. APK is now defined as "Assembly Polishing Kit" at first use in the text (line 209) and in the Supplementary Fig. 1 caption.

> *Replace "cost intense" with cost-intensive.*

Done.

> *"In accordance with to the Declaration of Helsinki" should be "in accordance with the Declaration of Helsinki".*

Fixed.

> *"Duplex-reads" should be Duplex reads.*

Fixed.

> *Define APK polishing kit at first use and describe what extra sequencing/data it requires.*

Done. "Assembly Polishing Kit (APK)" is defined at first use, and we now note that it requires one additional dedicated flow cell and a separate library preparation (lines 209 – 211).

> *In the Methods, "we first counted homopolymer-compressed k-mers… Then, we extract…" should be tense-consistent: then extracted.*

Fixed.

> *In the Pore-C methods, "happlotagging" appears to be a typo; use haplotagging.*

Fixed. We apologize for the typos and thank the reviewer for spotting them

> *"minimum binwidth of 1.000" likely means 1,000 bp. Please clarify.*

Done. All binwidth values have now been changed accordingly.

> *The HPRC comparison should specify whether HG002 was excluded in all relevant comparisons.*

HG002 is now included in the comparison plots, shown with a distinct point shape.

Plots now show both the HPRC-published HG002 and the T2T-ONT HG002 assembly generated with our method. We have adjusted the legends and state in the relevant Methods and Results parts if HG002 was included or excluded.

Reviewer #3

We thank the reviewer for the positive assessment and the comments that helped us to strengthen the evaluation of assembly quality.

Comment 1 : contiguity and multicopy genes

> *Contiguity: There has always been a disconnect between papers that present T2T genomes, and then a large fraction of missing multicopy genes. … A better metric is: how close are the ends of contigs to multicopy genes. Furthermore, the number of non-satellite DNA small contigs should be given.*

Done. We added the distance-to-multicopy-gene enrichment analysis and non-satellite small-contig counts to Supplementary Fig. 10, 11; Results (lines 176 - 183) and Discussion (lines 337 – 341).

We thank the reviewer for this precise characterization of the assembly contiguity problem. We performed the suggested analyses, and both support the reviewer's suggestion that multicopy genes and satellite regions are the principal sources of the remaining assembly fragmentation.

Satellite and non-satellite small contigs. We intersected the aligned reference span of each small contig (10 kb–10 Mb) with the CHM13v2.0 CenSat satellite annotation and classified a contig as satellite if ≥50% of its alignment overlapped annotated satellite sequence. Across the 23 assembled genomes and the HG002 benchmark assembly, we found a mean of 58 non-satellite small contigs per assembly (median 36, range 4–378; 1,381 in total), representing 58.2% of all small contigs (Supplementary Fig. 11). The remaining small contigs are classified as satellite contigs (mean 41 per genome, total 991, 41.8% of all small contigs) and map predominantly to centromeric and pericentromeric satellite arrays, where fragmentation is expected given the repeat-unit length relative to read length.

The non-satellite fraction is informative for the subsequent analysis of multi-copy-gene related assembly issues.

Contig ends and gaps at multicopy genes. To locate this remaining fragmentation, we identified the alignment boundaries of all haplotype-resolved contigs on CHM13v2.0 and measured the distance from each boundary to the nearest annotated multicopy gene locus. Across all assemblies, 22.2% of contig ends overlapped a multicopy gene locus directly and 41% fell within 10 kb, as did 68.5% of assembly gaps (Supplementary Fig. 10). This confirms the reviewer's suggestion: a substantial fraction of assembly breakpoints and gaps coincide with unresolved multicopy genes.

Taken together, the two analyses show that once satellite-driven fragmentation is set aside, the residual breaks are concentrated at multicopy gene loci rather than distributed through unique sequence. We have added both analyses to the Results (lines 176-183) and discussed multi-copy gene and satellite fragmentation as a limitation (lines 337 - 341).

Comment 1b : collapsed regions by satellite content

> *Collapsed regions (Fig 2C) should be separated out as satellite versus non-satellite DNA.*

Done. Flagged regions in Fig. 2C are now separated by repeat context (satellite, other repeats, non-repeat)., see Methods lines 491 – 494.

We have updated Fig. 2C so that collapsed regions are stratified into satellite, other repeats, and non-repeat sequence using the per-sample RepeatMasker annotation. As the reviewer anticipated, collapses fall predominantly in repetitive sequence.

Comment 2 : Mendelian inheritance as an orthogonal base-quality metric

> *Base quality. The k-mer validation of base quality can miss low complexity DNA where most errors are centered. Assessing violations of Mendelian inheritance in trios is an orthogonal approach. It will have some bias from misaligned sequences (make sure there are only two contigs mapped to any position where variants are evaluated for Mendelian inheritance).*

Done. Mendelian inheritance analysis added for both trios, restricted to positions with exactly two mapped contigs (Supplementary Table 3; Results lines 206 – 208; Methods lines 514 – 520).

We performed Mendelian inheritance analysis for both trio families (T2T00, T2T04). As suggested by the reviewer, we restricted evaluation to biallelic heterozygous child sites with exactly one assembly contig per allele (FORMAT/AD = 1,1), ensuring that misaligned sequences cannot inflate the error count. The raw Mendelian error rates are 4.0% and 3.6% for T2T00 and T2T04. To separate phasing artefacts from true base errors, we further classified each violation as either a switch/flip error (ALT allele present in at least one parent, consistent with haplotype-assignment errors) or a *de novo* error (ALT absent from both parents, indicative of a true assembly error or a genuine de novo mutation). Switch/flip errors account for 1.3–1.4 percentage points of the raw rate; the remaining de novo rate is 2.2–2.7%. Expressing the de novo error count relative to the diploid genome size yields a Mendelian QV of 48.8 and 49.5. These values are slightly above the corresponding Merqury k-mer QV (44.3 and 45.0) and are consistent with the reference-based evaluation against the GIAB HG002 benchmark. Results are reported in Supplementary Table 3.

Comment 3 : elevated chromosome 9 SV count

> *SVs - the high count of chr9 SVs likely reflect q-arm pericentromeric DNA where the reference does not match any sample well.*

Confirmed by intersecting chr9 SV calls with the pericentromeric heterochromatin annotation (Supplementary Fig. 20; Results lines 248 - 253).

We thank the reviewer for this observation and confirmed it with our data. We binned all chr9 SV calls in 2 Mb windows and intersected them with the CHM13v2.0 CenSat annotation, which delineates the pericentromeric heterochromatic block on the q-arm (chr9:40–81.7 Mb, spanning the p-arm/centromere transition, the centromere, and the large 9q heterochromatic block that GRCh38 represents only as N-gaps). The SV density in this region exceeds that of the chromosome arms by roughly 10–15 fold (Supplementary Fig. 20). Calls in this interval largely reflect differences between each sample's heterochromatic array and the CHM13 reference sequence rather than true inter-individual variation. We have added a few lines to the SV Results paragraph (lines 248 - 253) and included the per-chromosome density plot as Supplementary Figure 20.

Reviewer #4

We thank the reviewer for the thorough assessment of the manuscript and for the suggestion of an independent evaluation, which we have implemented by sequencing and assembling HG002.

Major comment 1 : independent evaluation

> *The evaluations based on the HERRO-corrected reads show good results, but some independent evaluation would be important. … e.g. by generating HiFi data for a limited number of samples, by applying this assembly recipe to a cell line for which independent assemblies are available (e.g. HG002), or by using public HPRC ONT data.*

Done. We generated a new HG002 assembly using our platform, which we evaluated against the GIAB HG002 benchmark (Supplementary Fig. 13, 14; Results lines 189-205, Methods lines 532 - 558).

We agree with the reviewer that metrics based on the same corrected reads used for assembly cannot detect errors introduced by the correction step. We therefore generated new sequencing data for the HG002 cell line using the same recipe as in the manuscript (three UL flow cells and one Pore-C flow cell), assembled it with the same pipeline, and compared the assembly to the GIAB HG002v1.1 benchmark using *GQC*. The GIAB benchmark set is independent of our data and therefore of the correction step.

The result does not indicate systematic errors from correction. Substitutions occur at roughly one transition and one transversion per megabase and make up only 2.6% of all discrepancies (QV 56.5). The remaining errors are indels in low-complexity sequence: 89.5% fall in homopolymer tracts and another 10.3% in other repeat classes. Non-repetitive sequence has only 781 errors in 4.83 Gb (QV 67.9) and coding sequence 56 errors in 70.3 Mb (QV 61.0). Continuity of the HG002 assembly follows the benchmark closely at scaffold level (Supplementary Fig. 14).

Our benchmark results reflect the known error patterns of Nanopore sequencing with R10.4 chemistry. Errors introduced by read correction would be expected to raise the substitution rate or to occur independently of sequence context, and we see neither. This analysis also led us to revise the QV values reported in the previous version of the manuscript (see the general response above and the answers to other reviewers).

Major comment 2: computational resource requirements

> *The authors emphasize the reduced wet-lab complexity of their approach, but I have the impression that this comes with the downside of a (much) higher demand for computational resources. … I didn't see a statistic on the compute resources and time required to perform the full assembly including the HERRO error-correction step. This should be added and, potentially, the authors might tone down the corresponding statement from the abstract (also in your discussion, L238).*

Done. Compute resource figure added (Supplementary Fig. 4), Results and Discussion updated, abstract and Discussion statements toned down.

We thank the reviewer for pointing at the lack of reporting compute time and resources. We have added Supplementary Fig. 4, which reports for all samples the runtime, RAM usage and CPU/GPU usage for the major steps of the workflow. We briefly discuss compute resources and runtime in Results (lines 135 - 139) and Discussion (lines 303 - 306).

We observed that compute requirements are moderate and do not represent a practical barrier to adoption. The main difference in compute runtime compared to HiFi-based workflows stems from two preprocessing steps: 1) super-accuracy basecalling (SUP) instead of high-accuracy basecalling (HAC) for all Nanopore reads. SUP basecalling can be performed in parallel to the flowcell run. 2) HERRO error correction, which required on median less than 24 hours (with a maximum of 2 days).

The hardware requirements are modest. The full workflow runs on a single high-memory, high-thread server (peak RAM below 250 GB) with at least one dedicated GPU for basecalling and HERRO. With the recent integration of HERRO into Dorado, error correction can also be performed directly on the PromethION during sequencing, completely removing the need for separate GPU servers and the extra compute time.

We have toned down the corresponding statements in the Introduction (lines 105 - 106) and Discussion (lines 303 – 306) as suggested, although we stand by our claim that the reduction in hands-on lab time is essential for efficient T2T generation and automation is much more impactful than a modestly increased compute time.

Comment: which HPRC assemblies were used

> *You are citing Logsdon et al., but it seems you are not comparing to their assemblies. It is somewhat unclear what HPRC assemblies you compare to. Are these assemblies released with Liao et al. or assemblies produced afterwards as part of the HPRC release 2?*

Done. We now specify the IDs of the HPRC assemblies used in the comparison in Supplementary Table 5 and Methods (479 – 484).

We apologize for the ambiguity. The 20 diploid assemblies used for comparison are from HPRC Release 2 (pre-release state), not from HGSVC3 (Logsdon et al.) or HPRC Release 1 (Liao et al.). The assembly index (assemblies_pre_release_v0.6.1.index.csv) was obtained from the HPRC intermediate assembly repository (github.com/human-pangenomics/hprc_intermediate_assembly, commit 41aa47d, 14 February 2025), predating the formal Release 2 announcement. We have added the full list of the 20 assemblies as Supplementary Table 5 and now specify the exact source in the Methods. We note that Logsdon et al. is cited only for the CenMap centromere-analysis methodology, not as an assembly comparison.

Minor comments

> *L77-79: This statement is slightly confusing because ref. 12 works from standard simplex ONT reads, i.e. something even less difficult to produce than ultra-long reads.*

Rephrased to: "whether the combination of ultra-long reads and Pore-C alone could yield high-quality diploid T2T assemblies at population scale, without HiFi data and without Duplex sequencing, had not been systematically demonstrated" (lines 79-82)

> *Figure 1B: White colour means "multiple contigs", which is not correct for Hap2 of chrY. Distinguish cases where a chromosome was assembled in multiple contigs vs was not present at all.*

Done. We added a distinct "Missing" category (muted grey) indicating that a chromosome is not present in the assembly, separate from the "multiple contigs" category, and updated both Fig. 2B and Supplementary Fig. 5.

> *L130ff: 95% can be quite strict for acrocentric chromosomes or chromosome Y, depending on its size variation (cf. Hallast et al., 2023). … can you add a supplementary figure that shows the total size of the T2T Y chromosomes.*

Done. T2T classification criterion clarified in Methods; chrY size figure added and Results extended (Supplementary Fig. 6; Methods lines 465 – 467 and 472 - 476, Results lines 154-156).

Our description of the 95% criterion was not clear enough, and we apologize for the confusion. The 95% is not a threshold on assembled chromosome length relative to the reference. We use *cov_cal* on the pairwise alignment of each assembly against T2T-CHM13. For each contig–chromosome pair, *cov_cal* builds a colinear alignment chain and requires two conditions at the same time: the chain must span at least 95% of the assembled sequence and at least 95% of the reference chromosome. The first ensures the sequence really corresponds to that chromosome, the second that the alignment reaches from one chromosome end to the other. Sequences passing this test are then called T2T contigs if they are gap-free, or T2T scaffolds if they still contain gaps.

The span is measured as the extent of the colinear chain, not as the number of aligned bases. Sequence present in one assembly but missing in the other therefore does not make the criterion fail, and a chromosome that is genuinely shorter than the reference can still be classified as T2T if it is assembled continuously from end to end.

The added Supplementary Fig. 6 shows this for chromosome Y. In the samples where chrY reached T2T status, assembled length ranges from about 42 Mb to 61 Mb, compared to 62.5 Mb in CHM13. This covers much of the known variation in Y length (Hallast et al., 2023). The shortest of them (T2T09, ~42 Mb) passes the same criterion as the longest and is classified as a scaffold only because it still has two gaps, not because of its length. We have kept the 95% criterion unchanged and clarified its description in the Methods (462 - 476).

> *L163ff: "k-mers generated from realignment": I do not understand how you obtain the k-mers from the re-alignment step.*

Fixed in Methods (lines 507 - 508).

We apologize that the wording was misleading: the k-mers are not derived from an alignment step. They are counted directly from the Illumina short-read data (>30× coverage per individual) and compared to the k-mers present in the assembled genomes, same as *Merqury* does. We have corrected the wording in the Results and Methods. This paragraph has also changed further: following the comments on independent evaluation, we now report reference-based QV against the GIAB HG002 benchmark as the primary accuracy metric and use the k-mer-based *Merqury* values only for relative comparison across the cohort (see the general response on QV).

> *L175: "Using Pore-C, assignment of paternal or maternal origin is, as expected, arbitrary": Adding a component for methylation-based parent-of-origin assignment would be a useful addition (along the lines of doi:10.1016/j.xgen.2022.100233).*

Done. Methylation-based parent-of-origin assignment implemented with PatMat and validated against trio phasing (Supplementary Fig. 15, 18; Results lines 226 – 236, Discussion lines 315 - 316).

We thank the reviewer for pointing us to this approach — it is very useful and led to a valuable addition to our computational workflow. Since Nanopore sequencing provides DNA methylation from the same reads used for assembly, parent-of-origin assignment can indeed be added without generating any additional data.

We implemented an adapted version of PatMat (Akbari et al., 2023). In the original workflow, chromosome-scale phase information comes from strand-seq data. In our case the assembly is already chromosome-scale phased through Pore-C, so we pass the assembly-derived phased VCF both as the per-variant input and as the chromosome-scale phase reference. PatMat then uses methylation at known imprinted DMRs in the haplotagged reads to determine which haplotype is maternal and which is paternal. The two inputs are independent: the phase blocks come from Pore-C, the parental labels from methylation, so the assignment is not circular.

We validated the assignment against trio-based phasing in both trios (T2T00 and T2T04). Methylation-based assignment reproduced the trio-derived parental origin in both cases (Supplementary Fig. 15). Across the full cohort, parent-of-origin was assigned for all but six autosomes (Supplementary Fig. 18). Sex chromosomes were not assigned, as no imprinted DMRs are annotated on chromosome X and X-chromosome inactivation would dominate the methylation signal. Final assemblies are now released with parent-of-origin-annotated haplotypes in PanSN naming. We have revised the statement in line 217 accordingly.

> *Fig 2A: The units are not fully clear to me. E.g. what does a negative fraction for MSC mean? What is the unit for the Completeness panel?*

Fixed. Corrected a plotting script error in Fig. 2A and Supplementary Fig. 7.

Thanks, this comment led us to an error in the plotting script, where a jitter value of 0.2 was incorrectly applied to the y-axis. After removing the jitter, the negative MSC values and genome-correctness values above 1 are resolved. This error mainly affected panels of metrics with fractional values (assembly covered, completeness, MMC, MSC, reference covered), while the impact of the error on panels with larger values was less pronounced (i.e. visually not recognizable, but still present).

> *L198ff: I think the term “continental groups” should be preferred over “superpopulation”. Also note that within the 1KG, the population labels represent self-identified ancestry, whereas the continental groups are not.*

Done. “superpopulation” replaced with “continental group” (line 259). Nature of the ancestry labels clarified (lines 379 - 381).

We have replaced “superpopulation” with “continental group” throughout the Results and in the Fig. 3A caption. We also clarify that the 1000 Genomes population labels represent self-identified ancestry, whereas the continental groups used for our cohort were assigned by us from reported country of birth and are therefore not self-identified (see also the response to the cohort-composition comment below).

> *L202: 50% strikes me as low-ish given a read length of a few kb for Pore-C (?); is that because of the read error rate?*

Done. Explanation added and per-sample statistics reported (Supplementary Table 4, Results lines 264 - 269). Phased-contact fraction added to the manuscript.

Read-level haplotype assignment ranges from 46% to 54% across the cohort. This rate is limited by the density of phased heterozygous SNVs overlapping short Pore-C fragments, not by the ONT error rate. In contrast to the published Dip3D approach, we do not call variants from the Pore-C reads themselves: we supply Dip3D with phased SNVs called directly from the phased assembly and use only its haplotagging and imputation steps, skipping the steps most sensitive to read errors.

Dip3D tags a fragment when it overlaps a phased heterozygous SNV, then imputes haplotypes to neighboring fragments on the same read (Chen et al. 2025). A read counts as assigned when at least one of its fragments carries a tag. Three factors set the ceiling: 1) SNV density: given human heterozygosity and the short length of Pore-C fragments, most fragments overlap no heterozygous SNV. Chen et al. report the same limitation (about 25% of monomers overlap a phased heterozygous SNV), 2) reference coverage: Dip3D requires read support at each supplied variant, so we restrict phasing to SNVs in regions with ≥5× uniquely mappable Pore-C depth, and about 28% of the reference falls below this threshold, mainly in repetitive and pericentromeric regions, 3) per-chromosome assignment: because Pore-C reads are concatemers spanning several chromosomes, a read is tagged on a given chromosome only if one of its fragments there overlaps a phased SNV. Empirically, about 21% of reads carry a tag per chromosome, and because each read spans several chromosomes the global rate rises to 46–54%.

Imputation raises the fraction of phased fragments per chromosome but not the number of phased reads, since it requires a SNV-tagged seed fragment within the same read. The read-level rate is therefore set by SNV coverage. Importantly, the read-level fraction understates the phasing information available for chromatin analysis: because assigned reads are longer and richer in contacts, 67–83% of contact pairs across the cohort come from haplotype-assigned reads, matching the 76% reported by Chen et al. (2025). These values are reported in Supplementary Table 4.

> *L300ff: please state the concrete composition of your cohort in sample numbers per ancestry. Also, Line 301 says "healthy adults" but later says "patients" and "Informed consent was obtained from all patients or their guardians". Could you clarify how recruitment was done?*

Done. Cohort composition by continental group and recruitment clarified. Participant metadata table added (Supplementary Table 1; see Methods: "Cohort description").

All participants are healthy adults who were informed about the study and gave written informed consent before inclusion. Recruitment was performed by open invitation at our institution. No patients or minors were included. The references to "patients" and to "consent by guardians" were drafting errors and have been removed. We apologize for this oversight.

We have added Supplementary Table 1 with sex, age band, and country of birth for each participant. The cohort comprises 23 participants including 17 single individuals and 2 trios. The continental-group counts are 18 Europe, 2 East Asia, 2 Americans, and 1 South Asia. The continental groups were assigned based on the country of birth reported by participants. We now state this explicitly in the Results, Methods, and the Fig. 3A caption.

> *L365: Can you clarify whether this corresponds to the method in ref 12 ("Hifiasm (ONT)")? Please include version numbers.*

Done. The reference has been corrected, and a machine-readable list of all tool versions has been added. Version numbers are now given throughout the Methods.

> *L386ff: please specify the versions used for the software tools (NucFlag, flagger, longphase, etc.). How did you create the RepeatMasker annotation (database used)?*

Done. Version numbers added for all tools, and the RepeatMasker run configuration and repeat library are now specified in the Methods.

> *393ff: how did you correct for the differences in overall genetic makeup between male and female samples?*

Done. No sex correction was applied. The sex-related difference in QC metrics is now reported explicitly, with the Y-bearing haplotype in male samples showing lower apparent completeness. Point shapes distinguishing male and female samples were added to Fig. 2A, with an explanatory sentence in the Results (lines 168-171) and figure legends.

> *404ff: I am unclear on how the alignment comes into play here? Merqury computes QV by comparing k-mers between assembled sequences and high-quality reads.*

Done. The reviewer is correct and this was also commented by Reviewer 2 (see answer above). Merqury estimates QV by k-mer comparison in reads without alignment. Our misleading wording has been corrected. Note that this paragraph has seen several updates to report the HG002-based QV as the primary metric.

> *Fig. S5: do you have a listing of the randomly selected HPRC assemblies?*

Done. The list of the 20 HPRC assemblies (from HPRC release 2) has been added as Supplementary Table 5 and the Methods section was amended (lines 479 – 484).

> *Fig. S6: I have trouble comparing the samples in these plots given that they only show absolute numbers/counts and not base pairs affected; please add another figure that enables that comparison.*

Done. Supplementary Fig. 8 (formerly S6) now shows the cumulative length of regions with issues in addition to counts, and the legend has been updated.

The count-based and length-based views show very similar between-sample patterns. The associated main-figure panel (Fig. 2C) was also updated at Reviewer 3's request to classify issues by repeat context.

> *Data Availability: The data is presently hosted at a research lab server. I suggest to host it at a public repository (e.g. ENA) so that it gets accessioned.*

Done. Data deposited in the German Human Genome-Phenome Archive (GHGA) under a persistent study accession.

We agree. The sequencing data and assembled genomes have been deposited in the German Human Genome-Phenome Archive (GHGA), which issues persistent accessions and is part of the federated EGA network. The study accession is GHGAS12079965883832. Because the data derive from identifiable human participants, access is controlled in line with our ethics approval and the participants' consent; the data access policy is available at the study page. The assembled genomes, together with annotations, also remain available on our public GitHub repository for convenient access. The Data Availability statement has been updated accordingly.

> *The authors link to a Cell Press competing interests form, which strikes me as odd when submitting to Nat Comms.*

Thanks for spotting this, it was a placeholder. We have replaced the link with a standard competing interest's statement.

Remarks on code availability

Done. The remarks on code availability overlap with those of Reviewer 1 and have been addressed there: we have added a versioned release tag, an MIT license, revised documentation, and step-by-step instructions for running the pipeline.